Tutorial 5 Solution

Problem 1

Consider the following sets of vectors. Determine (a) whether they are linear independent or linearly dependent, and (b) their spans.

(i) $S_1 = \{(1,-2),(2,-4)\}$;

(ii) $S_2 = \{(1,0),(0,1)\}$;

(iii) $S_3 = \{(0,1,0),(-2,2,0)\}$;

(iv) $S_4 = \{(0,1,0),(-2,2,0),(-2,3,0)\}$;

(v) $S_5 = \{(1,1,0),(0,-3,1),(2,4,-1)\}$.

Solution:

(i) We know from lecture slides that S_1 is linearly dependent; and

$$\text{span}(S_1) = \{a_1(1,-2) + a_2(2,-4) : a_1, a_2 \in \mathbb{R}\}.$$

Since $(2,-4) = 2(1,-2)$, we can simplify it to

$$\text{span}(S_1) = \{a_1(1,-2) + 2a_2(1,-2) : a_1, a_2 \in \mathbb{R}\} = \{(a_1 + 2a_2)(1,-2) : a_1, a_2 \in \mathbb{R}\}.$$

If we let $k = c_1 + 2c_2$, we can see that any vector in the span is just a scalar multiple of $(1,-2)$. This is precisely the line that passes through the origin and the point $(1,-2)$. In other words, the span is the set of all points (x,y) that satisfy the equation $y = -2x$.

(ii) To see that whether S_2 is linearly independent, consider

$$a_1(1,0) + a_2(0,1) = (0,0).$$

This is equivalent to $a_1 = 0$ and $a_2 = 0$, and therefore S_2 must be linearly independent. And its span is given by

$$\text{span}(S_2) = \{a_1(1,0) + a_2(0,1) : a_1, a_2 \in \mathbb{R}\} = \{(a_1, a_2) : a_1, a_2 \in \mathbb{R}\} = \mathbb{R}^2.$$

(iii) Again, we consider the equality

$$a_1(0,1,0) + a_2(-2,2,0) = (0,0,0).$$

Since the left-hand side is equivalent to $(-2a_2, a_1 + 2a_2, 0)$, the equality above is equivalent to $-2a_2 = 0$ *and* $a_1 + 2a_2 = 0$. Then it must be that $a_1 = a_2 = 0$, showing that S_3 is linearly independent.

For its span,

$$\text{span}(S_3) = \{a_1(0,1,0) + a_2(-2,2,0) : a_1, a_2 \in \mathbb{R}\} = \{(-2a_2, a_1 + 2a_2, 0) : a_1, a_2 \in \mathbb{R}\}.$$

Note that no matter what a_1 and a_2 we choose, the third entry is always zero. And any vector $(x, y, 0) \in \mathbb{R}^2$ can be written as $(-2a_2, a_1 + 2a_2)$ by letting $a_2 = -x/2$ and $a_1 = y + x$. Thus,

$$\text{span}(S_3) = \{(x, y, 0) : x, y \in \mathbb{R}\}.$$

Geometrically, the set of all vectors where the z-component is zero forms a flat, two-dimensional plane, sometimes called the xy-plane.

(iv) S_4 is linearly dependent because $(-2,3,0) = (0,1,0) + (-2,2,0)$, or equivalently,

$$(-2,3,0) - (0,1,0) - (-2,2,0) = (0,0,0).$$

By comparing S_3 in part (iii) and S_4, we see that $S_4 = S_3 \cup \{(-2,3,0)\}$. But since the vector $(-2,3,0)$ can be written as a linear combination of elements in S_3, it belongs to the span of S_3, and therefore including this vector doesn't enlarge the span. Therefore, $\text{span}(S_4) = \text{span}(S_3) = \{(x, y, 0) : x, y \in \mathbb{R}\}$.

(v) To determine whether S_5 is linearly independent, we consider

$$a_1(1,1,0) + a_2(0,-3,1) + a_3(2,4,-1) = (0,0,0).$$

This is equivalent to

$$\begin{aligned} a_1 + 2a_3 &= 0 \\ a_1 - 3a_2 + 4a_3 &= 0 \\ a_2 - a_3 &= 0 \end{aligned}$$

We see from the last equation that $a_2 = a_3$; plugging it to the second equation, we get $a_1 + a_3 = 0$. Now compare it with the first equation, we get $a_3 = 0$. Then because $a_2 = a_3$ and $a_1 + a_3 = 0$, it must be that $a_1 = a_2 = a_3 = 0$, showing that S_5 is linearly independent.

It is routine to see that

$$\begin{aligned} \text{span}(S_5) &= \{a_1(1,1,0) + a_2(0,-3,1) + a_3(2,4,-1) : a_1, a_2, a_3 \in \mathbb{R}\} \\ &= \{(a_1 + 2a_3, a_1 - 3a_2 + 4a_3, a_2 - a_3) : a_1, a_2, a_3 \in \mathbb{R}\}. \end{aligned}$$

One can verify that every vector $(x_1, x_2, x_3) \in \mathbb{R}^3$ can be written as $(a_1 + 2a_3, a_1 - 3a_2 + 4a_3, a_2 - a_3)$ by solving the system of equations

$$\begin{aligned} a_1 + 2a_3 &= x_1 \\ a_1 - 3a_2 + 4a_3 &= x_2 \\ a_2 - a_3 &= x_3 \end{aligned}$$

and getting

$$\begin{aligned} a_1 &= -x_1 + 2x_2 + 6x_3 \\ a_2 &= x_1 - x_2 - 2x_3 \\ a_2 &= x_1 - x_2 - 3x_3 \end{aligned}$$

Thus, $\text{span}(S_5) = \mathbb{R}^3$.

Problem 2

Determine the rank of the following matrices:

$$\boldsymbol{A} = \begin{pmatrix} 1 & 0 & 2 \\ 1 & -3 & 4 \\ 0 & 1 & -1 \end{pmatrix}, \qquad \boldsymbol{B} = \begin{pmatrix} 0 & 1 & 0 \\ -2 & 2 & 0 \\ -2 & 3 & 0 \end{pmatrix}, \quad \text{and} \quad \boldsymbol{C} = \begin{pmatrix} 2 & 0 & -1 & 3 \\ 1 & 1 & 2 & 2 \\ 2 & 0 & -1 & 1 \end{pmatrix}.$$

Solution: We consider the matrix $\boldsymbol{A}$ first. To calculate its rank, we need to find the rank of its columns (or, equivalently, the maximal number of its linearly independent columns), which is just $\text{rank}(S_5)$. Then because we showed in Problem 1 part (v) that in S_5, the maximal number of linearly independent vectors is three, we conclude that $\text{rank}\boldsymbol{A} = \text{rank}(S_5) = 3$.

For the matrix $\boldsymbol{B}$, note that its rank also equals to the rank of its rows (or, equivalently, the maximal number of its linearly independent rows), which is just $\text{rank}(S_4)$. Then because we showed in Problem 1 part (iv) that in S_4, the maximal number of linearly independent vectors is two, we conclude that $\text{rank}\boldsymbol{A} = \text{rank}(S_4) = 2$.

Finally, for the matrix $\boldsymbol{C}$, $\text{rank}\boldsymbol{C}$ equals to the rank of its rows, namely the rank of the set $C_R \equiv \{\boldsymbol{c}_1, \boldsymbol{c}_2, \boldsymbol{c}_3\}$, where $\boldsymbol{c}_1 = \begin{pmatrix} 2 & 0 & -1 & 3 \end{pmatrix}$, $\boldsymbol{c}_2 = \begin{pmatrix} 1 & 1 & 2 & 2 \end{pmatrix}$, and $\boldsymbol{c}_3 = \begin{pmatrix} 2 & 0 & -1 & 1 \end{pmatrix}$. To understand the number of maximal linearly independent vectors in C_R, consider the equation $a_1\boldsymbol{c}_1 + a_2\boldsymbol{c}_2 + a_3\boldsymbol{c}_3 = \boldsymbol{0}$, which is equivalent to the following four equations:

$$\begin{aligned} 2a_1 + a_2 + 2a_3 &= 0 \\ a_2 &= 0 \\ -a_1 + 2a_2 - a_3 &= 0 \\ 3a_1 + 2a_2 + a_3 &= 0 \end{aligned}$$

Plugging $c_2 = 0$ into the other three equations, we get

$$\begin{aligned} a_1 &= -a_3 \\ 3a_1 &= -a_3 \end{aligned}$$

then plug the first equation to the second, we obtain $-3a_3 = -a_3$, and therefore $-2a_3 = 0$, implying that $a_3 = 0$. Then because $a_1 = -a_3$, $a_1 = 0$. Thus, $a_1 = a_2 = a_3 = 0$, which means that C_R is linearly independent. Namely, the maximal number of linearly independent vectors in C_R is 3. Thus, $\text{rank}\boldsymbol{C} = \text{rank}(C_R) = 3$.

Problem 3

Let $\boldsymbol{A}$ be a $m \times n$ matrix, and $\boldsymbol{b} \in \mathbb{R}^m$. In the lecture we claimed that "there exists $\boldsymbol{x} \in \mathbb{R}^n$ such that $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$" is equivalent to "$\boldsymbol{b} \in \text{span}(\boldsymbol{A})$." Carefully argue why this assertion is true, and offer an example where $m = 2$ and $n = 3$.

Solution: First, let's analyze the statement "there exists $\boldsymbol{x} \in \mathbb{R}^n$ such that $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$." Let $\boldsymbol{A}$ be an $m \times n$ matrix. We can represent it by its columns: $\boldsymbol{A} = \begin{pmatrix} \boldsymbol{a}^1 & \boldsymbol{a}^2 & \cdots & \boldsymbol{a}^n \end{pmatrix}$, where

each column $\boldsymbol{a}^i$ is a vector in $\mathbb{R}^m$. Let x be a $(n \times 1)$ vector in $\mathbb{R}^n$:

$$\boldsymbol{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}.$$

By the definition of matrix-vector multiplication, the product $\boldsymbol{Ax}$ is a linear combination of the columns of $\boldsymbol{A}$ using the components of $\boldsymbol{x}$ as weights (or coefficients):

$$\boldsymbol{Ax} = x_1\boldsymbol{a}^1 + x_2\boldsymbol{a}^2 + \cdots + x_n\boldsymbol{a}^n.$$

Therefore, the equation $\boldsymbol{Ax} = \boldsymbol{b}$ is just a compact way of writing

$$x_1\boldsymbol{a}^1 + x_2\boldsymbol{a}^2 + \cdots + x_n\boldsymbol{a}^n = \boldsymbol{b},$$

implying that $\boldsymbol{b}$ *can be written as a linear combination of the columns of* $\boldsymbol{A}$.

Now let's turn to the statement "$\boldsymbol{b} \in \text{span}(\boldsymbol{A})$." Recall that $\text{span}(\boldsymbol{A})$ is defined as the span of the set of column vectors of $\boldsymbol{A}$, which we can write as $A_C := \{\boldsymbol{a}_1, \boldsymbol{a}_2, \ldots, \boldsymbol{a}_n\}$. By definition, this means that $\boldsymbol{b}$ can be expressed as a linear combination of vectors in A_C, or the columns of $\boldsymbol{A}$. But this is precisely the meaning of "there exists $\boldsymbol{x} \in \mathbb{R}^n$ such that $\boldsymbol{Ax} = \boldsymbol{b}$." The equivalence thus follows.

For the example, we can let

$$\boldsymbol{A} = \begin{pmatrix} 4 & 1 & -2 \\ 1 & 0 & 5 \end{pmatrix} \quad \text{and} \quad \boldsymbol{b} = \begin{pmatrix} -2 \\ 3 \end{pmatrix}.$$

Then "there exists $\boldsymbol{x} \in \mathbb{R}^n$ such that $\boldsymbol{Ax} = \boldsymbol{b}$" means that

$$\begin{pmatrix} 4 & 1 & -2 \\ 1 & 0 & 5 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -2 \\ 3 \end{pmatrix},$$

or equivalently

$$x_1 \begin{pmatrix} 4 \\ 1 \end{pmatrix} + x_2 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + x_3 \begin{pmatrix} -2 \\ 5 \end{pmatrix} = \begin{pmatrix} -2 \\ 3 \end{pmatrix}.$$

And "$\boldsymbol{b} \in \text{span}(\boldsymbol{A})$" means that $\boldsymbol{b}$ is in the span of the set of columns $\{(4,1),(1,0),(-2,5)\}$. That is, $\boldsymbol{b}$ can be written as a linear combination of the columns of $\boldsymbol{A}$:

$$x_1 \begin{pmatrix} 4 \\ 1 \end{pmatrix} + x_2 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + x_3 \begin{pmatrix} -2 \\ 5 \end{pmatrix} = \begin{pmatrix} -2 \\ 3 \end{pmatrix}.$$

Problem 4

Consider the system of equations

$$\begin{pmatrix} 4 & 1 & -2 \\ 1 & 0 & 5 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -2 \\ 3 \end{pmatrix}.$$

(1) Determine whether the system has a solution.

(2) (**Extra Practice Problem**) Determine whether the system has a solution, and if a solution exists, whether it has one or infinitely many solutions.

Solution:

(1) Take any vector $\boldsymbol{y} = (y_1, y_2)$, and let

$$\boldsymbol{yA} = \begin{pmatrix} y_1 & y_2 \end{pmatrix} \begin{pmatrix} 4 & 1 & -2 \\ 1 & 0 & 5 \end{pmatrix} = \begin{pmatrix} 4y_1 + y_2 & y_1 & -2y_1 + 5y_2 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \end{pmatrix}.$$

Then it must be that $y_1 = 0$, and plugging this into $4y_1 + y_2 = 0$, we have $y_2 = 0$. Consequently, $\boldsymbol{y} \cdot \boldsymbol{b} = 0 \cdot -2 + 0 \cdot 3 = 0$. Thus, whenever $\boldsymbol{yA} = 0$, it must be that $\boldsymbol{y} \cdot \boldsymbol{b} = 0$. The theorem on the slides then indicates that the system has a solution.

(2) We have

$$[\boldsymbol{A}|\boldsymbol{b}] = \begin{pmatrix} 4 & 1 & -2 & -2 \\ 1 & 0 & 5 & 3 \end{pmatrix};$$

it is routine to verify that $\text{rank}(\boldsymbol{A}) = \text{rank}([\boldsymbol{A}|\boldsymbol{b}]) = 2 < 3 = n$, and therefore by the theorem on the slides, the system of equations has infinitely many solutions.